

Diving Deeper into Volume Style Transfer

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Figure 1: NST Examples from Disney’s “Strange World”: Pando Thrower and Heart Healing Effects ©2023 Disney Enterprises, Inc.

ABSTRACT

The Pando Thrower is a weapon of great importance in Disney’s “Strange World.” The explosive Pando fuel mixture is expelled with turbulent, fluid-like motion, with branching arcs of lightning and the electrifying presence of high energy particles. During effects development, previous stylization methods were improved and expanded to enable the combination of distinctly different natural phenomena into a single fantastical effect. In this talk, we present our use of Neural Style Transfer (NST) to reshape volume simulations into lightning-laced electrical reactions.

CCS CONCEPTS

• Computing methodologies → Physical simulation; Machine learning; • Computer systems organization → Neural networks.

KEYWORDS

Physical simulation, Neural Style Transfer, Machine Learning

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1 INTRODUCTION

In the film, the Pando Thrower is a miracle invention the heroes create to protect their world. Our challenge was to design a high-energy stream of electromagnetic and chemical reactions consistent with the characteristics of its fuel source, the Pando plant. To

achieve this, the Pando Thrower’s continuous stream is created in two distinct steps. A fluid volume simulation, and its stylization by multiple NST passes. Since the Pando Thrower sim is based on the flamethrower rig developed for the Jaeger character, the focus of this effort is the application of NST to this effect, and its integration with the effects pipeline. This project is a continued collaboration with Disney Research Studios (DRS). Inspired by Neural Style Transfer (NST) methods for images [Gatys et al. 2015]. Our integration of TNST [Kim et al. 2019] employs a pre-trained Convolutional Neural Network (CNN) to extract high resolution shapes, textures, and features from 2D style images, and transfers them to the volume’s density field. In the following sections, we present these techniques, and our continued exploration of NST methods to expand artistic control for effects animators.

2 APPROACH

2.1 Volume Staging and Camera Motion

Since NST is based on a 2D image modification process [Gatys et al. 2015], there is some important asset preparation of both the source volume, as well as the shot camera. Our NST tooling for 3D volumes extends the functionality well beyond the original 2D image style transfer process, including a differentiable renderer built into its optimizer and support for multiple cameras. The optimizer also takes two volumes as inputs, the density and velocity fields. The optimizer uses the input velocity field to maintain temporal consistency, and the result is a modified density field, stylized in 3D world space from the point of view of at least one input camera.

Additionally, we export a style flow field, which is used to advect the stylized volume in Houdini as a post process. This approach opens up blending and volume shaping possibilities with access to the full range of Houdini’s toolset.

In some use cases, extreme or redundant camera motion can produce undesirable results. For this reason, we set up a custom

camera normalization pipeline. This will counter animate the camera to ensure its motion is minimized during style optimization. The camera animation will be subtracted from the volume's world space position, such that the volume is in the same relative position when viewed through the camera. This was essential for "Strange World" production shots, because the majority of Pando Thrower shots are set on a moving airship. While it's not uncommon to counter animate volume sources for fluid simulations, we needed a robust process to manage this preparation step. We built a custom camera tool with control for counter animation, based on reference geometry, as well as options to revert the counter-animated motion after NST optimization.

Additionally, the "Strange World" shot cameras were rendered with a fisheye lens, so we added a custom focal length offset on NST cameras to handle the overscan.

2.2 NST, Hyperparameters, and Style Tuning

Once the volume and cameras have been prepared, the NST process is mostly the same as 2D style transfer[Gatys et al. 2015]. A target style is specified and passed to the optimizer along with settings and the source inputs. Style layers, iterations, and learning rates are tuned in the same way specified in the TNST and Gatys et al papers.

However, a significant difference between 3D volume NST and 2D image NST is the importance of voxel-to-pixel ratio. For Pando Thrower effects that span multiple shots with continuity, maintaining a consistent voxel-to-pixel ratio is crucial to preserve a consistent look and scale of stylized features. To achieve this, we developed a custom Houdini tool that measures the voxel-to-pixel ratio. If we discovered that consecutive shots have the same volume effect at vastly different distances from the camera, the source volume may require resampling in one or more shots. The Pando Thrower shots averaged two pixels for each voxel in the density grid.

While this voxel-to-pixel ratio was important for a consistent look, it also provided useful feedback for general tuning of the stylized look, even when continuity was not critical. A one-to-one ratio resulted in finer, high-frequency shapes and textures, while a ratio of three or four pixels for each voxel in camera space produced larger style features, which is sometimes desirable.

When it comes to the art-direction of this process, precise control of input scale is critical. The size and resolution of the inputs greatly impact the scale and quality of transferred features. While the control setting in our NST tooling allows for adjustment of the voxel-to-pixel ratio during optimization, we discovered that better results were achieved by performing volume adjustments in Houdini in-between optimization passes.

2.3 Multiple NST Passes

In some situations, a single pass of NST may not achieve the desired artistic goals. In such cases, there were multiple passes of NST with different settings for octaves and other hyper-parameters for finer control of stylizations. By varying hyper-parameters and performing volume upsampling and density contrast adjustments

in-between multiple passes, we increased artistic control of the Pando Thrower effects in several shots.

2.4 Loss Graphs for Style Tuning

We created tools that allow us to measure the success of the style optimization by plotting loss value graphs on a per-octave, as well as a per-frame basis. We get the necessary feedback that allows us to tune the hyper parameters, most commonly the number of iterations and the learning rates. Unique learning rates for each of the octaves provided extra style control and fine tuning for all Pando Thrower shots.

2.5 Blending Multiple Styles

The original simulation and density source can be used to advect particles that match the density and velocity fields. The benefit to this, is the age attribute on the particles can then be rasterized to generate a custom mask for blending of multiple styles. When combined with the style flow field, this allows for greater control of stylization, both spatially and temporally. The mask is applied to each of the style flow fields, which are combined, and then applied to the source volume in a Houdini post-process. This avoids any obvious seams or artifacts that would otherwise be visible if the blending masks were applied directly to the density fields.

2.6 Computation and Process Distribution

NST computation is resource intensive, and it was critical to extend the effects pipeline to support a workflow that included new software libraries, new data formats, and a custom GPU mini farm that integrates with the existing pipeline. GPU processing is approximately 10x more performant and utilizes CUDA tools to process jobs. PyTorch-based NST jobs were set up and distributed from within a Houdini session, but remote jobs would compute NST without any Houdini dependency. NST typically requires all the available GPU memory, so limiting system overhead was critical. NVIDIA A6000 GPUs were used to process NST jobs on effects workstations as well as minifarm nodes.

3 CONCLUSIONS AND FUTURE WORK

Neural style transfer is a powerful technique that allows us to infuse animated visual effects with a desired artistic style. However, in order to achieve optimal results, it is crucial to properly prepare shot assets before applying NST. By assessing and modifying shot assets, including considerations for camera motion, lens distortion, voxel-to-pixel ratio, and volume placement, we can ensure that the NST process is performed with maximum efficiency and effectiveness. By following this workflow, we can achieve the desired results in animated visual effects and unlock the full potential of NST.

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